PROG1350A - Python III: Data Preprocessing and Statistical Analysis

Assignment II (Group Assignment)
Descriptive Statistics, Contingency Tables and Exploratory Data Analysis

Objective:

To use the attached customer churn dataset to perform descriptive statistics, frequency analysis, contingency-table analysis, and exploratory data analysis (EDA) using the Pandas techniques covered in Sessions 10 and 11. The expected output is a concise statistical profile of the customer base and evidence-based observations about customer churn.

Instructions:

This assignment is worth 15 points in total. To get 10 points, you must complete the required tasks listed below:

1. **Dataset overview and analysis plan (0.5 points)**

Load your cleaned dataset and report the number of rows and columns, identify the numerical and categorical variables, and briefly state which variables you selected for the statistical analysis and why.

2. **Descriptive statistics for numerical variables (1 point)**

Select four numerical variables from the cleaned dataset, including at least one variable related to customer tenure or charges. For each selected variable, calculate and display the mean, median, mode, range, variance, standard deviation, and IQR. Briefly identify where the mean and median differ noticeably and what this may suggest about the distribution.

3. **Compare churned and non-churned customers (2 points)**

Using Churn Label to define the groups, calculate the mean, median, and standard deviation for at least three numerical variables. Display the results in a clear table and identify important differences between customers who churned and customers who did not churn.

4. **Frequency and relative-frequency tables (2 points)**

Create frequency tables using value_counts() for Churn Label and at least three additional categorical variables. Create at least one relative-frequency table using normalize=True. Interpret the most common category and its proportion.

5. **Contingency tables and row percentages (2 points)**

Create contingency tables involving Churn Label and three categorical variables of your choice. Include at least a variable with three or more categories (for example, Contract Type or Payment Method). For each variable, two tables must be created: one table to show marginal frequencies using margins=True and a second version using row percentages with normalize='index'. Use the percentages to compare the churn distribution within the categories.

6. Exploratory data analysis findings (2 points)

Using only the statistics, frequency tables, and contingency tables produced in Tasks 2-5, identify at least three meaningful findings about the customer base or churn. Each finding must be supported by a specific numerical result or percentage. Do not use visualization methods*. Do not claim causation from descriptive or exploratory results.*

7. Final statistical summary (0.5 points)

Provide a concise summary of the main results. Include the final dimensions of the dataset used and summarize the most important patterns found in the analysis.

The remaining 5 points will be assigned according to your effort in explaining and justifying your analytical decisions and code in a “technical interview” of approximately 15 minutes (PowerPoint is not required, only Jupyter Notebook).

Focus your explanations on Tasks 1-6. You should be able to explain your code, why you selected particular variables, why specific descriptive measures are useful, how you interpreted differences between groups, why row percentages are appropriate for the contingency tables, and how your conclusions are supported by the results.

Area	**Points**
Variable/statistic selection and analytical decisions	1.0
Interpretation of central tendency and variability	1.0
Comparison of churn groups	1.0
Contingency-table interpretation	1.0
Overall EDA discussion and justification of conclusions	1.0
Total discussion/justification	5.0

Deliverables:

- A Jupyter Notebook file containing the code, tables, and **written explanations** for the statistical analysis.
- Code explanation / technical interview. All team members must participate; all members receive the same mark.

Note: ALL files need to be submitted to receive a mark. If a file is missing or does not compile (in case of the Jupyter Notebook file), the corresponding delay penalty will be applied (20% of the maximum value per day late).

Due Date:

- October 15, 11:59 PM

Teams:

- Groups of two

Dataset Description:

A fictitious dataset about churn from a Telecom provider. The variables describe customer status, demographics, contract information, subscription characteristics, service usage, and charges. The target variable for this assignment is Churn Label, which indicates whether the customer churned.

Relevant variables include Churn Label, Account Length (in months), Local Calls, Local Mins, Intl Calls, Intl Mins, Intl Active, Intl Plan, Extra International Charges, Customer Service Calls, Avg Monthly GB Download, Unlimited Data Plan, Extra Data Charges, State, Gender, Age, Under 30, Senior, Group, Number of Customers in Group, Device Protection & Online Backup, Contract Type, Payment Method, Monthly Charge, Total Charges, Churn Category, and Churn Reason.

Restrictions:

- Do not use visualization methods for this assignment. Do not include charts or plots.
- Use Pandas techniques covered in Sessions 10 and 11.